Generative AI Analysis and Ethics Report

CVE-Style Text Generation with DistilGPT2

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Overview

I fine-tuned DistilGPT2, a decoder-only Transformer, to generate continuations in the style of Common Vulnerabilities and Exposures (CVE) descriptions. The experiment used English descriptions published by CVE Numbering Authorities (CNAs) for CVE-2021 identifiers. Causal self-attention supports this task because the model can use the preceding context to predict the next token (Vaswani et al., 2017). The dataset is available at https://github.com/CVEProject/cvelistV5.

Validation perplexity, the exponential of mean validation loss, decreased from 58.69 to 12.35 after fine-tuning. The resulting continuations more often resembled advisories, but some repeated phrases, changed products, or made unsupported vulnerability claims.

Dataset or Prompt Description

I used the CVE List V5 snapshot at commit 3cb4c6bfb5fbd74de5b87f5158f97c0c56da8f88. The notebook reads the cves/2021 directory from a sparse checkout and writes data/cve_2021_descriptions.csv. Of 23,494 JSON files, 22,613 contained eligible English descriptions representing 178 CNAs. The median description was 39 words, the 75th percentile was 57, and the maximum was 696. The raw descriptions contained 1,466 exact duplicates beyond their first occurrence.

Repeated openings were common. “In the Linux kernel, the following” occurred 731 times. Of those records, 694 were published in 2024 and 37 in 2025, illustrating the difference between identifier year and publication year. For generation, I used three distinct four-word openings from held-out, human-authored records. Those openings were absent from the training set. The short, formulaic descriptions provide human-written examples of how advisories name products, versions, and impacts. The CVE Numbering Authorities authored the source descriptions, and redistribution is governed by the CVE Terms of Use (CVE Program, n.d.).

Model Design and Training Approach

DistilGPT2 is a distilled GPT-2 checkpoint (Hugging Face, n.d.). Radford et al. (2019) describe GPT-2, and Sanh et al. (2019) provide background on knowledge distillation for language models. DistilGPT2 uses six Transformer blocks and 12 attention heads. All

81,912,576 parameters were trainable. The compact model let me fine-tune the entire checkpoint on a 12 GB laptop GPU. I normalized whitespace, kept original punctuation, and removed short and duplicate descriptions. Grouping by the normalized first six words kept identical openings together to limit leakage of templated text between training and validation. This produced 16,618 training records and 4,202 validation records, or 20.2% of the cleaned corpus. The validation partition contained 2,569 opening groups.

Each record was encoded as one sequence capped at 256 tokens. An end-of-sequence token was added when space permitted, and padding was excluded from causal cross-entropy. I trained for three epochs in full precision with AdamW (Loshchilov & Hutter, 2019), a learning rate of 5×10^{-5}, batch size 16, and gradient norms clipped at 1.0. I weighted batch losses by the number of predicted non-padding tokens to obtain per-token epoch averages.

Output Evaluation and Interpretation

Validation loss fell in each epoch, with gains of 0.1412 and 0.0665 in epochs two and three. The validation-minus-training gap increased from 0.0523 to 0.1829 over those epochs. The widening gap and smaller validation improvement suggest possible emerging overfitting. Validation loss was still falling at the end of the run, so the experiment did not establish the optimal stopping epoch.

Epoch	Training loss	Validation loss
1	3.0225	2.7214
2	2.5279	2.5802
3	2.3308	2.5137

The two checkpoints used matched prompts and seeds under two temperature and top-p settings, with 48 new tokens allowed per continuation. Top-p sampling restricts each draw to the smallest set of likely tokens whose cumulative probability reaches the chosen threshold (Holtzman et al., 2020). The baseline sample baseline-1-2-2 drifted into a sports score and photo credit. The matched sample fine-tuned-1-2-2 used a component and version range, then described a questionable causal link between a segmentation fault and a buffer overflow. Sample fine-tuned-1-1-1 repeated its description of upload functionality. Sample fine-tuned-2-1-2 paired an 8.x product label with a cutoff of 7.1.6.

Distinct-n measures the share of observed word sequences of length n that are unique (Li et al., 2016). The baseline reached distinct-2 of 1.00 at temperature 0.7 and top-p 0.90 while averaging only 6.17 words per output and frequently emitting blank lines. The baseline also repeated a sentence in baseline-2-2-1. These are examples of degeneration in this run, a broader problem studied by Holtzman et al. (2020). The fine-tuned checkpoint had distinct-1 scores of 0.61 and 0.71 across the two settings. Each score represents just six outputs, and temperature and top-p changed together, so this comparison cannot isolate the effect of either setting.

Ethical Considerations and Responsible Use

Samples fine-tuned-3-1-1 and fine-tuned-3-2-1 switched from a mobile-device-management prompt to claims about Intel products. Their distinct five-gram overlap with the training corpus was 0.963 and 0.933, respectively. Research on training-data extraction shows that language models can reproduce training material (Carlini et al., 2021). Here the metric measures phrase reuse across the corpus and cannot identify whether either sample copied a single record.

Advisory phrasing and real vendor names can make unsupported claims appear credible. The corpus contains hundreds of records with identical openings, and fine-tuned-3-1-1 illustrates how learned advisory phrasing can carry a real vendor name into an unrelated context. If treated as a genuine advisory, a generated continuation could contaminate a vulnerability feed or divert patching and investigation resources. This scenario illustrates the broader misinformation and misuse risks identified by Weidinger et al. (2021). I labeled the continuations as fictional and used them to evaluate the model. Operational claims about affected products, versions, or impact should be checked against the official CVE record and vendor advisory.

Limitations and Future Improvements

The evaluation covered three prompts, 24 outputs, and one training run. Frequent templates and uneven representation across CNAs may give the model more exposure to some advisory styles than others. Opening-group splits can still allow paraphrased templates to appear in both partitions. Long source records were truncated at 256 tokens, and the experiment used no formal factual-accuracy measure. A follow-up could evaluate more prompts across several seeds, vary one decoding parameter at a time, and use validation-based checkpoint selection. Checking vendor and product claims against official records would also help distinguish factual errors from stylistic ones.

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